Lab 03: Real-World Data Linear Regression

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BSCE – 3D

A. Data

Data Source

Two World Bank indicators were used:

- Population, total — Indicator Code: SP.POP.TOTL
- GDP (current US$) — Indicator Code: NY.GDP.MKTP.CD

GDP source:
https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=PH

Population source:
https://data.worldbank.org/indicator/SP.POP.TOTL?locations=PH

Description of the Data

The dataset represents the annual population and GDP of the Philippines from 2011 to 2025. Population represents the total number of people in the country, while GDP represents the country's gross domestic product measured in current US dollars.

For the linear regression analysis, population is used as the independent variable $x$and GDP is used as the dependent variable y.

There are 15 paired observations in the dataset.

Variables

Variable	Symbol	Unit	Role
Population, total	x	people	Independent variable
GDP, current US$	y	US dollars	Dependent variable

The regression model used in the laboratory is:

$$\boxed{y = a_0 + a_1 x}$$

where $a_0$is the intercept and $a_1$is the slope.

Data Used

The following 15 paired observations were used in the regression analysis:

Year	Population (people)	GDP (US$)
2011	98,248,614	234,216,872,955
2012	100,175,512	261,920,481,725
2013	102,076,336	283,902,818,577
2014	103,767,130	297,483,582,902
2015	105,312,992	306,445,117,723
2016	106,735,719	318,626,957,647
2017	108,119,693	328,480,739,673
2018	109,465,287	346,841,689,750
2019	110,804,683	376,823,091,728
2020	112,081,264	361,750,638,591
2021	113,100,950	394,087,419,150
2022	113,964,338	404,353,550,708
2023	114,891,199	437,055,646,226
2024	115,843,670	461,671,537,274
2025	116,786,962	487,086,070,094

B. Python Script

```
# ============================================================
# LAB 03 - REAL-WORLD DATA LINEAR REGRESSION
# Numerical Methods - Section 3D
#
# Dataset:
# Philippines Population vs GDP (2011-2025)
# Source: World Bank
# ============================================================
```

```
import numpy as np

import matplotlib.pyplot as plt


# ------------------------------------------------------------

# 1. DATA

# ------------------------------------------------------------

# x = Population (people)

# y = GDP (current US$)


years = np.array([

    2011, 2012, 2013, 2014, 2015,

    2016, 2017, 2018, 2019, 2020,

    2021, 2022, 2023, 2024, 2025

])


population = np.array([

     98248614,

    100175512,

    102076336,

    103767130,

    105312992,

    106735719,

    108119693,

    109465287,

    110804683,

    112081264,

    113100950,

    113964338,
```

```
    114891199,
    115843670,
    116786962
], dtype=float)

gdp = np.array([
    234216872955,
    261920481725,
    283902818577,
    297483582902,
    306445117723,
    318626957647,
    328480739673,
    346841689750,
    376823091728,
    361750638591,
    394087419150,
    404353550708,
    437055646226,
    461671537274,
    487086070094
], dtype=float)

# ----------------------------------------------------------
# 2. LEAST-SQUARES LINEAR REGRESSION
# ----------------------------------------------------------
# Regression model:
#
# y = a0 + a1*x
```

```
#
# Slope:
# a1 = [n Σ(xy) - Σx Σy] / [n Σ(x²) - (Σx)²]
#
# Intercept:
# a0 = y_mean - a1*x_mean

n = len(x := population)
y = gdp

sum_x = np.sum(x)
sum_y = np.sum(y)
sum_x2 = np.sum(x ** 2)
sum_xy = np.sum(x * y)

a1 = (n * sum_xy - sum_x * sum_y) / (
    n * sum_x2 - sum_x ** 2
)

x_mean = np.mean(x)
y_mean = np.mean(y)

a0 = y_mean - a1 * x_mean

# -----------------------------------------------------------
# 3. CALCULATE PREDICTED VALUES AND RESIDUALS
# -----------------------------------------------------------

y_pred = a0 + a1 * x
```

```
residuals = y - y_pred

# ------------------------------------------------------------
# 4. SUM OF SQUARED ERRORS (SSE)
# ------------------------------------------------------------

SSE = np.sum(residuals ** 2)

# ------------------------------------------------------------
# 5. R-SQUARED
# ------------------------------------------------------------

SST = np.sum((y - y_mean) ** 2)

r_squared = 1 - (SSE / SST)

# ------------------------------------------------------------
# 6. STANDARD ERROR OF ESTIMATE
# ------------------------------------------------------------

standard_error = np.sqrt(SSE / (n - 2))

# ------------------------------------------------------------
# 7. PREDICTION
# ------------------------------------------------------------
# We choose 120,000,000 people.
# This value is NOT included in our 2011-2025 dataset.
```

```
prediction_population = 120_000_000

predicted_gdp = a0 + a1 * prediction_population

# ------------------------------------------------------------
# 8. DISPLAY RESULTS
# ------------------------------------------------------------

print("=" * 65)
print("LAB 03 - REAL-WORLD DATA LINEAR REGRESSION")
print("=" * 65)

print("\nDATASET")
print("-" * 65)
print("Source: World Bank")
print("Country: Philippines")
print("Years: 2011-2025")
print("Number of observations:", n)

print("\nVARIABLES")
print("-" * 65)
print("Independent variable (x): Population (people)")
print("Dependent variable (y): GDP (current US$)")

print("\nREGRESSION RESULTS")
print("-" * 65)

print(f"Intercept (a0): {a0:,.6f}")
print(f"Slope (a1):     {a1:,.6f}")
```

```
print("\nRegression equation:")
print(f"y = {a0:,.2f} + {a1:,.6f}x")

print("\nERROR AND FIT STATISTICS")
print("-" * 65)
print(f"SSE (Sr):         {SSE:,.2f}")
print(f"R-squared (r²):    {r_squared:.6f}")
print(f"R-squared (%):     {r_squared * 100:.2f}%")
print(f"Standard error:    {standard_error:,.2f}")

print("\nPREDICTION")
print("-" * 65)
print(f"Population (x):    {prediction_population:,.0f} people")
print(f"Predicted GDP:     ${predicted_gdp:,.2f}")

print("\nINTERPRETATION")
print("-" * 65)
print(
    "The positive slope indicates that population and GDP have "
    "a positive linear relationship in this dataset."
)

print(
    f"The R-squared value of {r_squared:.4f} means that approximately "
    f"{r_squared * 100:.2f}% of the variation in GDP is explained "
    "by the linear regression model."
)
```

```
print(
    "The residuals represent the difference between the actual GDP "
    "and the GDP predicted by the regression line."
)

print("=" * 65)

# ------------------------------------------------------------
# 9. PRINT DATA TABLE
# ------------------------------------------------------------

print("\nDATA USED")
print("-" * 65)
print(f"{'Year':<8}{'Population':>18}{'GDP (US$)':>25}")

for i in range(n):
    print(
        f"{years[i]:<8}"
        f"{population[i]:>18,.0f}"
        f"{gdp[i]:>25,.2f}"
    )

# ------------------------------------------------------------
# 10. GRAPH: DATA AND FITTED LINE
# ------------------------------------------------------------

# Sort x values so the fitted line is displayed smoothly.
sort_index = np.argsort(x)
```

```
x_sorted = x[sort_index]
y_pred_sorted = y_pred[sort_index]

plt.figure(figsize=(10, 6))

plt.scatter(
    x,
    y,
    label="Actual data"
)

plt.plot(
    x_sorted,
    y_pred_sorted,
    linewidth=2,
    label="Least-squares regression line"
)

plt.xlabel("Population (people)")
plt.ylabel("GDP (current US$)")
plt.title("Philippines Population vs GDP (2011-2025)")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()

plt.savefig(
    "lab03_regression_graph.png",
    dpi=300
)
```

```
plt.show()

# ------------------------------------------------------------
# 11. RESIDUAL PLOT
# ------------------------------------------------------------

plt.figure(figsize=(10, 6))

plt.scatter(
    x,
    residuals
)

plt.axhline(
    0,
    linewidth=1.5
)

plt.xlabel("Population (people)")
plt.ylabel("Residual (US$)")
plt.title("Residual Plot - Philippines GDP Regression")
plt.grid(True, alpha=0.3)
plt.tight_layout()

plt.savefig(
    "lab03_residual_plot.png",
    dpi=300
)
```

```
plt.show()


# ----------------------------------------------------------

# END OF PROGRAM

# ----------------------------------------------------------
```

C. Regression Results

The fitted straight-line model is:

$$y = a_0 + a_1 x$$

Using the calculated least-squares coefficients:

$$a_0 = -971{,}396{,}052{,}833.99$$
$$a_1 = 12{,}180.948548$$

Therefore, the regression equation is:

$$\boxed{y = -971{,}396{,}052{,}833.99 + 12{,}180.948548x}$$

where:

- x= population in people
- y= GDP in US dollars

Slope

$$\boxed{a_1 = 12{,}180.948548}$$

The positive slope indicates that population and GDP have a positive linear relationship in the selected dataset.

In practical terms, the fitted model estimates an increase of approximately US$12,180.95 in GDP for every additional person in population within the mathematical relationship represented by this dataset.

However, this does not mean that adding one person directly causes GDP to increase by that amount. The regression only describes the relationship observed in the data.

Intercept

$$\boxed{a_0 = -971{,}396{,}052{,}833.99}$$

The intercept represents the model's predicted GDP when population is zero.

Since a population of zero is outside the range of the actual dataset, the intercept has little practical meaning here. It is mainly a coefficient needed to define the regression line.

Sum of Squared Errors

The calculated sum of squared errors is:

$$\boxed{S_r = SSE = 4{,}255{,}011{,}561{,}737{,}829{,}220{,}352}$$

The SSE represents the total squared difference between the actual GDP values and the GDP values predicted by the regression line.

A smaller SSE generally indicates smaller prediction errors for the observations being modeled.

Coefficient of Determination

The calculated coefficient of determination is:

$$\boxed{r^2 = 0.944065}$$

or:

$$\boxed{r^2 = 94.41\%}$$

This means that approximately 94.41% of the variation in GDP in the selected dataset is explained by the fitted linear regression model using population as the independent variable.

The remaining approximately 5.59% is not explained by this simple linear model.

Standard Error

The standard error of the estimate is:

$$\boxed{s_{y/x} = 18{,}091{,}671{,}610.78 \text{ US\$}}$$

This represents the typical size of the prediction errors of GDP around the fitted regression line, expressed in US dollars.

Regression Graph

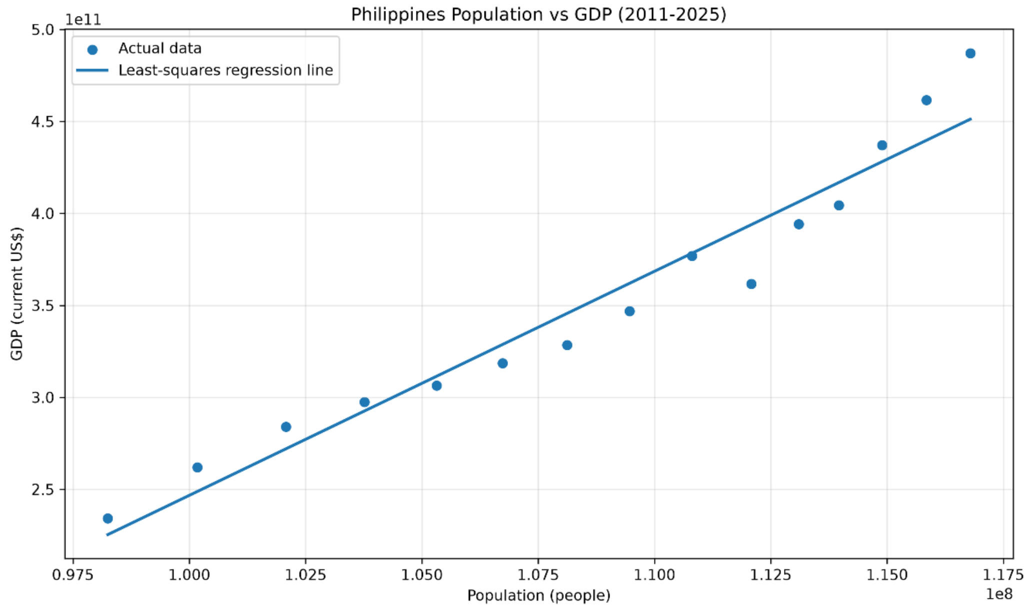

Figure 1. Philippines Population vs. GDP with Fitted Regression Line, 2011–2025

The graph shows the actual population-GDP observations together with the fitted least-squares regression line. The points generally follow an upward trend, which is consistent with the positive slope obtained from the regression.

Residual Plot

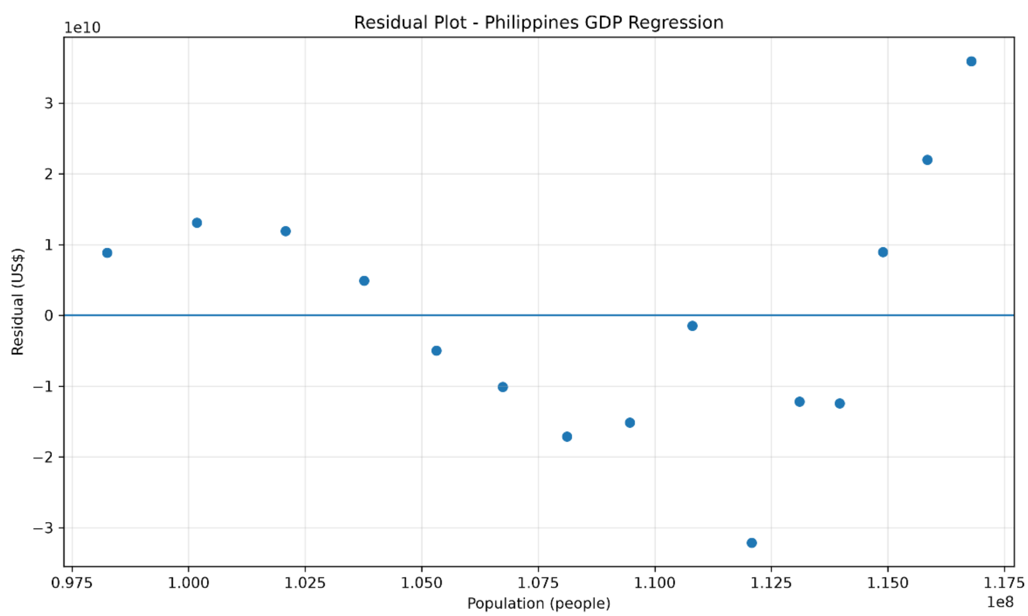

Figure 2. Residual Plot of the Philippines GDP Regression

The residuals represent the difference between the actual GDP values and the GDP values predicted by the regression equation.

The residual plot can be used to see how the prediction errors are distributed around zero. The residuals are not exactly zero because the regression line does not pass through every data point.

Prediction

For the prediction, the selected population value is:

$$x = 120{,}000{,}000 \text{ people}$$

Using:

$$y = -971{,}396{,}052{,}833.99 + 12{,}180.948548x$$

Substitute:

$$y = -971{,}396{,}052{,}833.99 + 12{,}180.948548(120,000,000)$$

Therefore:

$$\boxed{y = 490{,}317{,}772{,}891.94 \text{ US\$}}$$

or approximately:

$$\boxed{\$490.32 \text{ billion}}$$

Based on the fitted linear regression model, if the population of the Philippines reaches 120 million people, the predicted GDP would be approximately US$490.32 billion.

Interpretation

The regression analysis shows a positive linear relationship between population and GDP for the Philippines from 2011 to 2025. The slope of 12,180.948548 indicates that GDP generally increases as population increases within the selected data.

The coefficient of determination, $r^2 = 0.944065$, indicates that approximately 94.41% of the variation in GDP is explained by the fitted linear model. This suggests that the line represents the overall trend of the selected observations fairly well.

The standard error of approximately US$18.09 billion indicates the typical magnitude of the differences between the observed GDP values and the values estimated by the regression model.

The residuals show the prediction errors of the model. In particular, the 2020 observation differs noticeably from the general upward trend, so the regression line does not perfectly represent every observation.

Overall, the linear regression provides a useful mathematical description of the relationship between population and GDP in the selected dataset. However, GDP is influenced by many factors, so population alone should not be interpreted as the sole cause of changes in GDP.